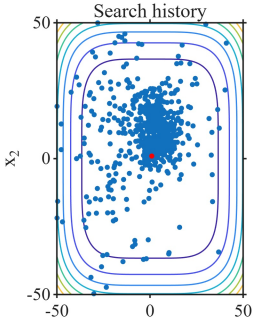
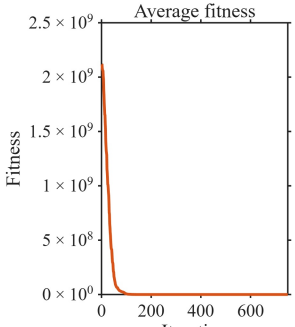
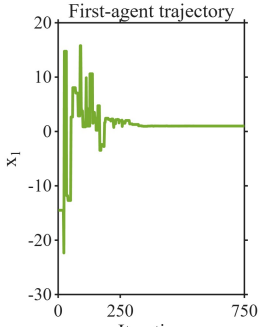
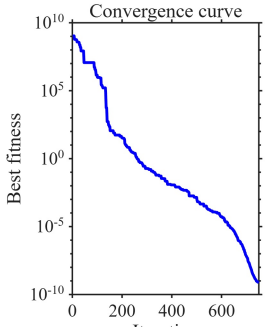

F13
Fitness
12
10
8
6
4
2
5
0
-5
x2
-5
0
5
x1


Search history
50
0
-50
x2
-50
0
50
x1


Average fitness
2.5 × 10^9
2 × 10^9
1.5 × 10^9
1 × 10^9
5 × 10^8
0 × 10^0
Fitness
0
200
400
600
Iteration


First-agent trajectory
20
10
0
-10
-20
-30
x1
0
250
750
Iteration


Convergence curve
10^10
10^5
10^0
10^-5
10^-10
Best fitness
0
200
400
600
Iteration